MILITARY SPECIFICATION

HOSE, RUBBER, FLAME-RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hose for use in fuel and oil lines in aircraft engine installations, tanks, and other combat vehicles.

* 1.2.1 Part number. Specification part number for items described in this specification will be formulated as shown in Section 6.3.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-S-735	-	Standard Test Fluids, Hydrocarbon.

MILITARY

* MIL-P-775	-	Hose, Hose Assemblies, Rubber, Plastic, Fabric or Metal (including Tubing, and Fittings, Nozzles, and Strainers) Packaging of.
MIL-L-6082	-	Lubricating Oil; Aircraft Reciprocating Engine (Piston).
MIL-M-6002	-	Marking; Standard Hose, Aircraft.

STANDARDS

FEDERAL

* FED-STD-601	-	Rubber, Sampling and Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, SA-ALC, Kelly Air Force Base, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MS33658 - Fitting End, Hose Connection, Standard Dimension for.

* MIL-STD-129 - Marking for Shipment and Storage.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

NAVY DEPARTMENT

General Specifications for Inspection of Material

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The component parts of this hose shall be a continuously extruded butadiene-acrylonitrile tube, plies of reinforcement, and a chloroprene outer cover.

3.2 Design.

3.2.1 Hose diameter. The inside and outside diameter of the hose shall conform to the dimensions given in Table I.

TABLE I. Inside and Outside Diameter

Nominal Inside Diameter (inch)	Outside Diameter (inch)		Nominal Inside Diameter (inch)	Outside Diameter (inch)	
	Minimum	Maximum		Minimum	Maximum
3/16	9/16	21/32	7/8	1-5/16	1-13/32
1/4	5/8	23/32	1	1-7/16	1-17/32
5/16	23/32	13/16	1-1/8	1-9/16	1-11/16
3/8	25/32	7/8	1-1/4	1-11/16	1-13/16
7/16	7/8	31/32	1-3/8	1-13/16	1-15/16
1/2	15/16	1-1/32	1-1/2	1-31/32	2-3/32
5/8	1-1/16	1-5/32	1-3/4	2-7/32	2-11/32
3/4	1-3/16	1-9/32	2	2-15/32	2-19/32

3.2.2 Diameter tolerance. The tolerance on the inside diameter shall be ±1/64 inch for sizes 3/8 inch inside diameter or less, and ±1/32 inch for all other sizes; the outside diameter tolerance shall be as specified in Table I.

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3.3 Hose construction.

3.3.1 Inner tube. The tube of the hose shall be a seamless, continuous extrusion of butadiene-acrylonitrile copolymer compounded with the necessary ingredients to meet the physical requirements of this specification. It shall have a smooth bore, and shall be clean and free from pitting.

3.3.2 Reinforcement. The reinforcement shall consist of two braided or helically wound plies one of which shall be asbestos. The braid angle shall be such that the finished hose meets the physical requirements of this specification.

3.3.3 Outer cover. The outer cover shall be of a chloroprene polymer suitably compounded to meet the requirements of this specification. It shall be of uniform thickness and free from pitting.

3.4 Installation. The force required to insert a corresponding size fitting conforming to MS33658 into the hose shall be in accordance with Table II.

TABLE II. Installation Test

Nominal Inside Diameter (inch)	Maximum Force (pounds)
3/16 - 7/16	35
1/2 - 1	75
1-1/8 - 2	125

3.5 Thickness. The thickness of the tube shall be not less than 0.030 inch.

3.6 Tensile strength.

3.6.1 The original tensile strength of the tube shall be at least 1,250 psi; the original tensile strength of the cover shall be not less than 1,000 psi.

3.6.2 After immersion in fuel conforming to Specification TT-S-735, Type III, the tensile strength of the tube shall be not less than 500 psi.

3.6.3 After immersion in fuel conforming to Specification TT-S-735, Type I, the tensile strength of the cover shall be not less than 500 psi.

3.6.4 After immersion in oil conforming to Specification MIL-L-6082, grade 1100, the tensile strength of the tube and cover shall be not less than 70 percent of the original tensile strength.

3.7 Elongation.

3.7.1 The original elongation of the tube shall be at least 250 percent, and of the cover, at least 200 percent.

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3.7.2 After immersion in oil, the tube shall have an elongation of at 50 percent; the elongation of the cover shall be at least 100 percent.

3.8 Adhesion.

3.8.1 The adhesion between the tube and braid shall be at lease 10 pounds per inch. The adhesion between all other parts shall be at least 8 pounds per inch.

3.8.2 After boiling in fuel conforming to Specification TT-S-735, Type I, the adhesion between the tube and braid shall be at least 8 pounds per inch.

3.9 Burst pressure. The burst pressure of the hose shall conform to Table III.

TABLE III. Burst Pressure of Hose

Size of Hose (Inside Diameter)	Burst Pressure (Minimum psi)
3/16 - 1/2 inch	600
5/8 - 1 inch	500
1-1/8 - 1-1/2 inch	400
1-3/4 - 2 inch	350

3.10 Volume swell. The volume swell of the tube after immersion in fuel conforming to Specification TT-S-735, Type III, shall not exceed 50 percent. The volume swell of the cover after immersion in fuel conforming to Specification TT-S-735, Type I, shall not exceed 35 percent.

3.11 Low temperature flexibility. When cooled to -40°F the hose shall not break nor show visible cracks in the tube or cover.

3.12 Flame resistance. The hose shall not leak or burst when subjected to the Flame test specified in pragraph 4.4.

3.13 Lengths. Unless otherwise specified, the hose shall be furnished in 10-foot lengths except that 10 percent of the order may be furnished in random lengths over 3 feet. All ends shall be cut squarely. When specified, hose shall be furnished in uncut lengths up to and including 50 feet. A tolerance of ±10 percent will be permitted on the length, except that pieces less than 3 feet long shall not be furnished.

3.14 Hose shall be no more than 12 months old from date of manufacture to date of delivery.

3.15 Identification of product. The hose shall be legibly marked along the longitudinal axis with a broken red stripe and a broken white stripe. The stripes shall conform to Specification MIL-M-6002, and shall show size, date of manufacture, and Specification MIL-H-7938. This information shall be repeated at least every 6 inches.

3.16 Workmanship. Workmanship shall be of the quality necessary to produce hose free from all defects which affect proper functioning in service.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Navy department inspection. When procurement is inspected under the supervision of the Navy Department, the general inspection procedures shall be in accordance with the Navy Department General Specifications for Inspection of Material.

4.2 Examination of product. All hose shall be carefully examined in order to insure conformance to the requirements of this specification with respect to materials, construction, size, marking, and workmanship.

4.3 Sampling tests. An 8-foot sample shall be selected from each lot of 10,000 feet or fraction thereof, and subjected to the following tests. If production is in excess of 10,000 feet per day, one sample may be selected each day. A lot shall consist of hose of the same cure submitted for inspection at the same time.

4.3.1 Thickness. A specimen of tube shall be buffed just to the point where the fabric or other impression disappears. The thickness shall be measured with a micrometer graduated to 0.001 inch under a load of 3 $\pm$0.1 ounces exerted by a weight through a flat contact foot 0.25 $\pm$.01 inch in diameter.

4.3.2 Tensile strength. The tensile strength of the tube and cover shall be determined as outlined in FED-STD-601.

4.3.3 Elongation. The elongation of the tube and cover shall be determined as outlined in FED-STD-601.

4.3.4 Fuel immersion. Dumbbell and volume swell specimens described in FED-STD -601 shall be cut from the tube and cover. The cover specimens shall be immersed in fuel conforming to Specification TT-S-735, Type I, for 48 hours at room temperature. The tube specimens shall be immersed in fuel conforming to Specification TT-S-735, Type III, for 48 hours at room temperature. Tensile strength tests and volume swell shall be determined at room temperature within 5 minutes after removal from the fuel described in FED-STD-601. Tensile strength calculations shall be based on the cross sectional area before immersion.

4.3.5 Oil immersion. Dumbbell specimens as described in FED-STD-601 shall be cut from the tube and cover, the thickness measured, and the specimen immersed in oil conforming to Specification MIL-L-6082, Grade 1100, at a temperature of 250° $\pm$2°F for a period of 70 hours. The container shall be fitted with a reflux condenser and the specimens shall be supported at least 1/2 inch above the bottom of the container. After immersion, the specimens shall be wiped with acetone, allowed to stand at room temperature for 24 hours, and tested.

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4.3.6 Adhesion. The Adhesion test shall be conducted in accordance with FED-STD-601.

4.3.6.1 Boil test. A 4-inch length of hose shall be boiled for 1 hour (using a reflux condenser) in fuel conforming to Specification TT-S-735, Type I. The fuel and specimens shall then be allowed to cool for 24 hours. Adhesion determinations shall be made within 30 minutes after removal of the specimens from the fuel.

4.3.7 Low temperature flexibility.

4.3.7.1 Hose 3/4 inch inside diameter and under. A straight piece of hose 12 inches long shall be conditioned at -40°F for 5 hours. The hose shall then be flexed around a mandrel or cylinder having a diameter equal to 10 times the outside diameter of the hose being tested and which has also been conditioned for 5 hours at -40°F.

4.3.7.2 Hose over 3/4 inch inside diameter. Three 4-1/2 by 1/4-inch specimens of the tube and cover stock buffed to a thickness of 0.075 ±0.010 inch shall be placed in a bending device consisting of two parallel jaws of suitable length, 2-1/2 inches apart, designed to hold the strip specimens in a loop position. The specimens shall be firmly fastened to the jaws and extended at least 3/4 inch into each jaw. The jaws shall be supported in guides so that they may be rapidly brought together from 2-1/2 inch to 1 inch separation. The rubber specimens, mounted in this apparatus, shall be exposed to cold dry air at -40°F for 5 hours, and while still in the cold chamber, the jaws shall be rapidly moved from 2-1/2 inch to 1 inch separation. The specimens shall then be examined for signs of cracking.

4.4 Special test. One sample of hose of a new design, size of sample to be 1 inch inside diameter, shall be subjected to the following test. This test need not be run on other hose sizes of similar construction or reorders of an approved design provided no changes have been made in the hose construction.

4.4.1 Flame resistance. One end of a 2-foot length of hose shall be attached to a source which supplies oil conforming to Specification MIL-L-6082, grade 1100, under a pressure of 60 ±2 psi. The other end of the hose shall be attached to a discharge line equipped with a valve so adjusted that the oil flow is equal to five times the square of the nominal inside diameter of the hose, expressed in gallons per minute. A flame shall be placed under the center of the hose and so adjusted that the temperature of the cover of the hose is 2,000° ±50°F. A period of 3 minutes shall be allowed for adjusting the flame and measuring temperature. The flame shall be permitted to remain constant for 5 additional minutes. Temperature readings shall be made at the end of each minute by means of a potentiometer and thermocouple. The end of the thermocouple shall touch the cover of the hose at the hottest point. (The presence of a carbon tetrachloride fire extinguisher during the test is recommended).

4.5 Rejection and retest. Rejected hose shall not be resubmitted for approval without full particulars concerning previous rejection and measures taken to overcome the defects.

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* 5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (6.2), in accordance with MIL-P-775.

* 5.2 Packing. Packing shall be Level A, B, or C as specified (6.2), in accordance with MIL-P-775.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The hose covered by this specification is intended for use in aromatic fuel and oil lines in aircraft engine compartments, tanks, and other combat vehicles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following: Applicable levels of preservation, packaging and packing required (see 5.1 and 5.2). State size and length in accordance with provisions of paragraph 6.3.

6.3 Definitive specification part number. The specification part number is a definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selecting (from the requirements) options available in this specification as follows:

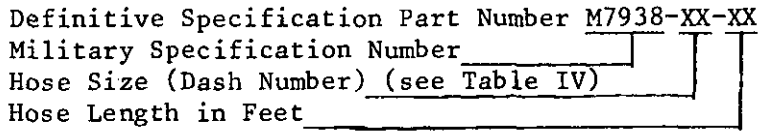


TABLE IV. Hose size designators

-3=3/16=.1875	-14=7/8=.875
-4=1/4=.25	-16=1=1.00
-5=5/16=.3125	-18=1 1/8=1.125
-6=3/8=.375	-20=1 1/4=1.25
-7=7/16=.4375	-22=1 3/8=1.375
-8=1/2=.5	-24=1 1/2=1.50
-10=5/8=.625	-28=1 3/4=1.75
-12=3/4=.75	-32=2=2.00

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6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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